Muhammad Abdullah Khan

AI Engineer | Backend Developer

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SUMMARY

Software Engineer and Applied AI Developer focused on fine-tuning open-source models, building local RAG pipelines, and designing backend API infrastructure. Experienced in serving models locally using vLLM and llama.cpp, implementing agentic tool-calling workflows, and writing reliable Python backends.

PROFESSIONAL EXPERIENCE

FlyRank AI, *Backend AI Engineer Intern* — 05/2026 – 06/2026 | Remote - Chicago, Illinois

- Built a local model serving system using vLLM, enabling the backend to automatically swap out custom-trained models on a single GPU to minimize hardware costs.
- Created an intelligent query router that automatically directs user requests to SQL databases, vector search indexes, or web APIs depending on user intent
- Automated the data ingestion pipeline by building an agent that parses raw PDFs and structures the data into vector databases for fast retrieval.

QUBIC, *Web3 QA Analyst* — 11/2023 – 03/2024 | Remote - Dubai, UAE

- Wrote automated Python scripts to test system integrations and APIs, reducing manual regression testing cycles from 2 days to under 3 hours.
- Created and executed load-testing scripts across 50+ Web3 endpoints to locate and document critical system bottlenecks.

EDUCATION

Bachelor of Science in Software Engineering (BSSE), — 09/2023 – Graduation: 09/2027

COMSATS University Islamabad — Lahore, Pakistan

Academic Standing: CGPA: 3.45 / 4.00

Relevant Coursework: Machine Learning, Pattern Recognition, Data Science, Software Engineering Principles, Software Architecture, Data Structures & Algorithms, Database Systems.

SKILLS

Agentic AI & Orchestration
Graph-Based Agents (LangGraph), Multi-Agent Coordination, Memory Management, Model Context Protocol (MCP), Tool Routing, NeMo Guardrails.

Machine Learning & NLP
PEFT/LoRA, Model Quantization (GGUF), Semantic RAG, Vector DBs (LanceDB, ChromaDB), AST Parsing (Tree-sitter), PyTorch, Data Preprocessing, Llama.cpp/ Ollama Serving

Backend & Infrastructure
Python, FastAPI, PostgreSQL, RESTful APIs, Docker, Git/GitHub, CI/CD Automation, Linux Systems.

Agentic Workflows
Claude Code, Google Antigravity, OpenAI Codex, Agentic SDLC Integration, Parallel Code Generation.

PROJECTS

OmniVLA, *Hardware-Constrained Autonomous GUI Agent* — 01/2026 – Present

- Built a system using LangGraph that coordinates multiple AI agents to automatically navigate and control computer desktop interfaces.
- Fine-tuned a lightweight vision model to accurately identify and click on specific user interface buttons with high pixel precision.
- Stored agent execution histories in a local database (LanceDB) to save context, compressing the model to run efficiently on small local GPUs.

OpaqueCI, *AI-Powered Zero-Trust Code Review Agent* — 05/2026 – 06/2026

- Developed an automated code-review assistant that triggers via FastAPI webhooks to audit GitHub pull requests for flaws before code is merged.
- Used Tree-sitter to analyze code structures and variable definitions, cutting down false-alarm security warnings by over 150 alerts per scan.
- Built a fast local search index in LanceDB to match submitted code against corporate safety standards in under 1.5 seconds.

VigilAI Proxy, *LLM Security Gateway & Telemetry* — 04/2026 – 05/2026

- Built an API gateway using FastAPI to securely route requests to different AI providers (like OpenAI and Claude) with minimal latency overhead
- Wrote middleware to sanitize user inputs, block malicious prompts, and redact sensitive personal data before forwarding requests to model APIs.
- Leveraged AI-driven coding assistants (Claude Code, Google Antigravity) to rapidly prototype and launch a dashboard showing gateway performance metrics.